

PSY 205L QUANTITATIVE RESEARCH METHODS & STATISTICS FOR PSYCHOLOGICAL SCIENCE 2

Summary In this two-semester course series (204L/205L) we will learn how to turn curiosity into knowledge. We will focus on the design of research studies and the use of statistical methods to become better consumers, creators, and communicators of psychology and human neuroscience research. By the end of this semester, students will have the foundational methodological and statistical knowledge and skills to conduct their own independent research.

After taking this class, you will...

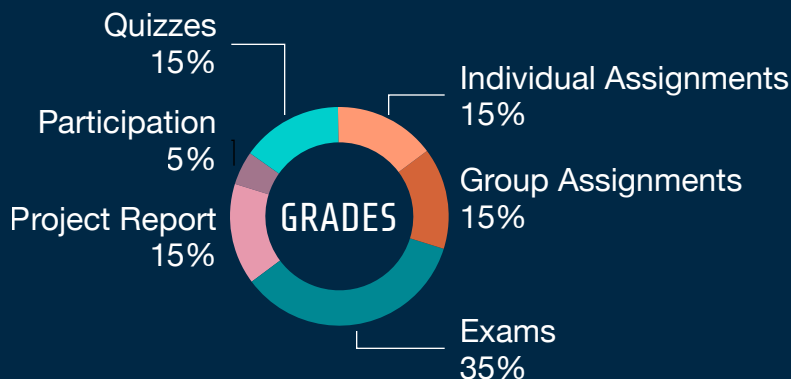
Know current best practices for research study design and statistical analysis

Have learned how to do basic statistical analysis of quantitative data

Be able to communicate scientific research results effectively and accurately

Be skilled in critically evaluating scientific research results & claims

Understand ethics in research and how to conduct ethical science



LECTURE TTh 1:25-2:40pm, Physics 128
SECTION T3:20p (03), T4:55p (04), W8:45a (06), W12p (05), W1:40p (07)

NO BOOK: all materials in Canvas

PROJECT: teams in sections

TWO MIDTERMS: 2/17 & 4/16 in class

OPTIONAL FINAL: 5/2 2-5pm

WEEKLY ASSIGNMENTS: Sunday 10pm

FREE SOFTWARE: <https://jasp-stats.org/>

GR Samanez-Larkin, Ph.D.

g.samanezlarkin@duke.edu

<http://bit.ly/BookProfSL>



Jade Terry

[email](#)

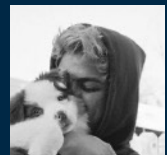
T3:20, T4:55



Jaime Rios

[email](#)

W12, W1:40



Max Freudenheim

[email](#)

W8:45



Teaching team

Duke SPRING 2026

This semester (**PSY205L**) is focused on experimental research design, statistical tests for simple and factorial designs (e.g., t-tests, regression, ANOVA, and interactions), how to use statistical software, and science communication.

Overview of Course Components and Grading

This course combines lectures with credit/no-credit **quizzes (15% of grade)**, interactive activities during normal class periods, sections/labs with a teaching assistant, weekly **assignments (40% of grade)** often with both group-based and individual components, an in-class **midterm exam (20% of grade)**, an **optional cumulative final exam (will replace midterm if higher grade)**, and a group-based **research report based on data collected during the semester (20%)**. **Participation credit (5% of grade)** will be given for attending section/lecture and providing responses to in-class surveys. Details below on each course component.

TEXTBOOK & SOFTWARE

TEXTBOOK: There is no required textbook for this course. You paid for this class to learn from local Duke experts so all materials will be provided either via Canvas or via lectures and sections/labs. Some students learn better from reading a text and we completely understand that. If you feel like it would be helpful to have additional reading material, we will provide suggested materials (chapters from relevant textbooks) for each week on Canvas. Reading these materials is optional.

SOFTWARE: There are many different software options for basic data analysis. We will use the free and open source software, **JASP**, for demonstrations and activities during lecture and lab sections. This is not a programming course and we assume no prior background in programming or statistics. JASP does not require any programming. We will teach you the basics in JASP in a non-scary and supportive way. Our focus will be on teaching critical statistical thinking more than developing a foundation for programming.

LECTURES AND QUIZZES



There are two lecture periods each week where live lectures may be given that blend the presentation of information with student-instructor and student-student interaction. Each classroom lecture will have a credit/no-credit quiz (15% of course grade) of 1-3 multiple choice questions. Some content will be provided in pre-recorded lectures segments. Each recorded mini-lecture will have credit/no-credit quiz (15% of course grade) questions. Some of these lecture class periods will be dedicated to group meetings with the instructors.

LABS/SECTIONS

In addition to the lecture components, students will attend a weekly lab session. During these sessions, students will review lecture material with a teaching assistant and have more TA-supported hands-on practice with that week's material in preparation for the weekly assignment.

PARTICIPATION

Participation credit is given for attending sections/labs as well as contributing to data collection throughout the semester. Some weeks there will be short Qualtrics surveys to collect data from students. These data will be collected, anonymized, and compiled by the teaching team and analyzed by the students for assignments, lab activities, and/or exams. Lecture attendance will be collected via Acadly.

WEEKLY ASSIGNMENTS

There will be weekly assignments posted on Canvas (and in Gradescope). Assignments are mini-projects aimed at reinforcing concepts discussed in class and developing your independent methodological and statistical thinking and skills. Some assignments are done in small groups (to build your research project report described below) and others should be done on your own. For the parts to be completed on your own, you may feel free to discuss those sections with other students before you start doing the actual work of the assignment, but the finished work that you turn in must be your own.

Group assignments will be weighted by your contributions. Group members will rate everyone in their group on contributions in the middle and at the end of the



semester. **Group assignments will be retroactively weighted by the average of the anonymous group contribution scores for the first and second half of the semester.**

Assignments are due by Sunday at 10pm Eastern time. Late assignments will lose 20% for every day they are late. The maximum grade for turning in the late assignment before 10pm on Tuesday is 80%, before 10pm on Wednesday is 60%, before 10pm on Thursday is 40%, and before 10pm on Friday is 20%. No assignments will be accepted after Friday. **All assignments count (we don't drop the lowest).**

RESEARCH REPORT

You will complete a research study proposal in small groups of ~5 (selected from classmates in your assigned sections/labs) over the course of the semester. This proposal will be in the form of a Short Report which is a scientific manuscript format that contains a brief background and rationale for a study with hypotheses (Introduction), the detailed methods of the study (Method), fully reported statistical results with relevant tables and figures (Results), and a brief description of how the study contributes new knowledge, an interpretation of the findings, and the implications for future research / policy change / interventions (Discussion).

You will not have the opportunity to procrastinate on this report. Many of your weekly assignments will involve drafting sections of this document. You will be rewarded throughout the semester for making consistent progress on this project. At the end, you'll compile everything and feel great! **Our goal is to build to a very satisfying and non-stressful end-of-semester where you get to just bask in the glory of your new knowledge and skill.** At the end of the semester, your group will present your study and results to the class during lecture. **The group project grade will also be weighted by the contributions of team members.**

MIDTERM EXAMS

There will be two in-class paper-and-pencil midterm examinations (short answer) that involve evaluating research methods and practical statistical knowledge. The exam will cover the topics discussed in lecture and sections/labs up until that date in

205L including the topics from 204L. These are non-traditional exams where you will creatively address potential research methods concerns and reason through statistical output (graphs and inferential claims). You must be present to complete the exam and turn in a paper exam at the end of class. If you are entitled to exam accommodations via the Student Disability Access Office (SDAO), you may complete exams at the testing center. **Exams are closed book/materials. No electronic devices can be used for these exams.** Make-up exams will be given only under exceptional circumstances. You must contact the instructors before the exam if an emergency arises.

OPTIONAL FINAL EXAM

There will be an optional cumulative final examination during the exam period that is

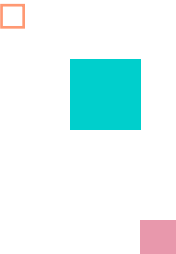
Our course policy is as follows: We don't care when you learn it; all that matters is that you learn it!

a longer version but the same format as the midterm exams. The final exam will cover the topics from the entire semester of PSY205L as well as PSY204L. Make-up exams will not be given since this exam will be optional.

In this spirit, there are no risks to taking the optional final - only potential benefits. If you don't take it, your grade will not be affected. If you take it and the grade is higher than the average of your midterm exam grades, both of your midterm exam grades will be replaced by your final exam grade. You do not need to request this; it will happen automatically. If your grade on the final exam is lower than the average of your midterms, your midterm exam grades will not be replaced by your final exam grade. If the final exam grade is lower, we will not include that score anywhere in your final course grade.

GRADE SCALE

The letter-grade scale for this course is:



A+ = 99-100%	A = 93-98%	A- = 90-92%
B+ = 87-89%	B = 83-86%	B- = 80-82%
C+ = 77-79%	C = 73-76%	C- = 70-72%
D+ = 67-69%	D = 63-66%	D- = 60-62%
F = <60%		

Grades will be rounded up, for example, from 94.50% to 95%.

PSYCH MAJOR, PRE-MED REQUIREMENTS, QS

Completion of both semesters of the course are required to meet the statistics and methods requirements for Psychology. Both semesters are also required to meet the pre-med statistics requirement. If you or a friend are not a Psychology major and just looking for a QS, you can take 204L and be done. Of course, we are happy to have you stay for 205L, too! This is the semester when things get really fun.

BEHAVIOR AND COMMUNITY STANDARDS



Duke Community Standard

Duke University is a community dedicated to scholarship, leadership, and service and to the principles of honesty, fairness, respect, and accountability. Members of this community commit to reflect upon and uphold these principles in all academic and nonacademic endeavors, and to protect and promote a culture of integrity. Duke University has high expectations for students' scholarship and conduct. In accepting admission, students indicate their willingness to subscribe to and be governed by the rules and regulations of the university, which flow from the Duke Community Standard (DCS).

Regardless of course delivery format, it is the responsibility of all students to understand and follow all Duke policies, including academic integrity (e.g., completing one's own work, following proper citation of sources, adhering to guidance around



group work projects, and more). Ignoring these requirements is a violation of the DCS.

Students can direct any questions or concerns regarding academic integrity to the Office of Student Conduct and Community Standards at conduct@duke.edu and can access the DCS guide at <https://dukecommunitystandard.students.duke.edu/>.

MENTAL HEALTH AND WELL-BEING

Duke is committed to holistic student wellbeing; this includes one's mental, emotional and physical health. The university offers resources to help students manage daily stress, to encourage intentional self-care, and access just-in-time support. If you find you need support, your mental and/or emotional health concerns are impacting your day-to-day activities, your academic performance, or you need someone to talk to, the resources below are available to you:

- **DukeReach**. DukeReach provides comprehensive outreach services to support students in managing all aspects of wellbeing, including referrals, and follow-up services for students who are experiencing significant challenges related to mental health, physical health, social adjustment, and/or a variety of other stressors. You can reach the DukeReach team at dukereach@duke.edu.
- **Counseling and Psychological Services (CAPS)**, (919) 660-1000 CAPS services include individual and group counseling services, psychiatric services, and workshops. CAPS also provides referrals to off-campus resources for specialized care.
- **TimelyCare** (formerly known as Blue Devils Care) is an online platform that is a convenient, confidential, and free way for Duke students to receive 24/7 mental health support through TalkNow and scheduled counseling. <https://Timelycare.com/bluedevils>
- **BC Fellows for Healthy Relationships** meet with students individually and in groups, supporting the development of healthy relationships and building meaningful community in all areas of a student's lives. To get in touch with the BC Fellows, go to <https://students.duke.edu/wellness/duwell/sexualhealth/balthrop-cassidy-fellowship/>.

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- **DukeLine**: Students who want to anonymously connect with a Peer Coach can text 984-230-4888 from 5-11 pm daily. DukeLine offers in-the-moment anonymous, non-emergency text support from a peer. Additional information is available at <https://sites.duke.edu/dukeline>.
 - **DuWell**, (919) 681-8421, provides Moments of Mindfulness (stress management and resilience building) and meditation programming (Koru workshop) to assist students in developing a daily emotional wellbeing practice. All are welcome, and no experience is necessary. For addition information visit <https://students.duke.edu/wellness/duwell/>.

ACADEMIC SUPPORT, ACCESSIBILITY, & ACCOMMODATION

The **Academic Resource Center** (the ARC) offers services to support students academically during their undergraduate careers at Duke. The ARC can provide support with time management, academic skills and strategies, course-specific tutoring, and more. ARC services are available free to any Duke undergraduate student studying any discipline. You can contact the Academic Resource Center by phone at (919) 684-5917, by email at theARC@duke.edu, or by visiting arc.duke.edu.

The **TWP Writing Studio** provides attentive, non-evaluative readers who can help students see whether they're expressing ideas in ways that are clear to someone outside our class. Consults with Writing Studio can be productive at any stage in a student's process, including before writing has begun. Visit <http://twp.duke.edu/twp-writing-studio> to learn more and to schedule an in-person or online appointment.

Course Materials Costs. This course shouldn't cost you one penny. We don't use a textbook in this class. If you are having difficulty with the textbook and/or supply costs or access to materials associated with this or any other course, here are some resources:

- **Karsh Office of Undergraduate Financial Support**: Regardless of your aid package, Karsh offers loans and resources for connecting students with campus programs that might help alleviate course costs.

- **DukeLIFE**: The Course Material Assistance program offers assistance for eligible students. Students who are eligible for DukeLIFE benefits are notified prior to the start of the semester; program resources are limited.
- **Duke Libraries**: The library's Top Textbook Program allows you to rent some textbook for 3 hours at a time.
- **Duke Link**: You can reserve rental equipment from the Link for course-specific technology needs, such as a Digital Voice Recorder, HD Video Camera, TI-84 Plus CE, DSLR camera kit, Tripod, Shotgun Mic, iPad Mini 4, Handheld Projector, or GoPro.
- Highly-aided students who have limited access to computers may request loaner laptops through the **LIFE Loaner Laptop Program**. Please note that supplies are limited.

Academic Accommodations. If you are a student with a disability and need accommodations for this class, it is your responsibility to register with the **Student Disability Access Office (SDAO)** and provide them with documentation of your disability. SDAO will work with you to determine what accommodations are appropriate for your situation. Please note that accommodations are not retroactive, and disability accommodations cannot be provided until a Faculty Accommodation Letter has been given to me. Please contact SDAO for more information: sdao@duke.edu or access.duke.edu.

Religious Accommodations. University policy permits students to be absent from class to observe a religious holiday. Accordingly, Trinity College of Arts & Sciences and the Pratt School of Engineering have established procedures for students to notify their instructors of an absence necessitated by the observance of a religious holiday. Please submit requests for religious accommodations at the beginning of the semester so we can work to make suitable arrangements well ahead of time. You can find the policy and relevant notification forms for Trinity College ([Religious Observance & Holidays](#)) and Pratt School ([Religious Observance & Holidays](#)).

COURSE POLICIES/EXPECTATIONS



Inclement Weather Policy. In the event of inclement weather or other related events that prohibit class attendance, I will notify you how we will make up missed course content and work. Asynchronous virtual catch-up methods may apply.

Flexible Attendance & Due Date Policies. Responsibility for class attendance rests with individual students. Since regular class attendance is expected, students must accept the consequences of failure to attend. If you miss a lecture or section, you will likely need to get notes from someone else (or watch the lecture recording). There will also be many interactive activities during lecture and especially during section periods. If you miss a lecture, there is nothing specific you need to do to make up these activities. Try them out on your own. They are purely for your own benefit to support your learning. If you need to miss a section, you will need to notify the TA in advance and ask for details on how to make up that section which will likely involve some coordination with your group members.

Artificial Intelligence (AI) Policy. Two important objectives of this class are helping students develop their analytical/critical thinking skills and their scientific writing skills, prioritizing accuracy and sound reasoning. Developing these skills will strengthen your learning and prepare you for many competitive workplaces where you can't get by with AI alone. Given these objectives, students in this class are approved to use AI or generative AI tools in limited ways, as follows. (If you have questions, ask the teaching team):

Approved Use #1: Searching for relevant research articles to find and read, such as with consensus.app. Note that many tools, such as ChatGPT, may generate fake (non-existent) or irrelevant references. ChatGPT regularly includes references that do not actually have evidence for a claim Chat makes based on that reference.

Approved Use #2: Developing outlines (to organize ideas that you have generated). To evaluate the flow and consistency of various sections of text assembled together (for example, when combining revised sections of text in your project report).

Approved Use #3: Doing basic editing of spelling and grammar at the word- or sentence-level (e.g., using Microsoft Editor or Grammarly's grammar checker).



Approved Use #4: Formatting citations per APA style. However, please note that some generative AI tools don't get all of these details correct, either.

Approved Use #5: To confirm procedures for running statistical tests in JASP and to compare your answers to statistics questions to what an AI might generate. Note that these answers may be inaccurate and/or are unlikely to be formatted to the same standards that we expect in this course.

Approved Use #6: To generate initial drafts of slides/presentations. Make sure these are edited to the standards of the course and you are prepared to deliver the presentation orally and answer questions from your peers and the teaching team.

All other uses are not permitted.

Minor edits of ChatGPT output does not count as your own writing. Completely AI-generated submissions are not permitted, even if edited, and we will make referrals to the Office of Student Conduct and Community Standards in the event of suspected violations of the AI policy for this class.

Finally, it is your responsibility to verify that anything AI-generated (see approved uses) is confirmed to be accurate, and you also will be asked to make AI disclosures on homework assignments, as well as your final research study proposals.

Assignment due dates are fixed. No assignments get dropped. However, in recognition of possible extra personal and academic stress and health concerns this semester, **we will grant a one-time no-questions-asked assignment extension of 24 hours.** More details on Trinity attendance policies are available [here](#). More details on Pratt attendance policies are available [here](#).

Assignment/exam/report regrades: The assignments and exams are written by the professors who also write grading guides but the actual grading is primarily done by the TAs. If you have questions about a grade, please first contact your TA to discuss the assignment. If there is a grading error, they can fix it. If it was graded correctly but you are contesting the grading, you will need to request a regrade by the professors. One of us will re-grade the entire assignment. Your grade may go up, down, or not change. This is not meant to be threatening. If you ask for our extra



time through a re-grade, we will do the best job we can re-grading it to be as fair as possible. We reserve the right to deny regrade requests if they become too frequent. We are committed to providing you the best learning experience possible and that takes time. We understand how critical your grades are. Know that everything we do leading up to that grade is intended to prepare you for success.

This is a collaboration: We all have the same goal. We are deeply invested in this course and your success in it. Every member of this teaching team loves teaching this content. We will do our best to get to know each and every one of you. Please bring your whole self to this class. Creating a welcoming and inclusive classroom experience together as instructors and students is an important priority for all of us. We are not therapists but we can direct you to one if you need it (see above). We are here to help you not only learn the content of this course but become confident!

If you are ever confused or struggling, please come to office hours or reach out to us. It is our job to support your learning experience here at Duke. Please do not keep your struggle to yourself.



There is no shame in having a hard time. All of us on the teaching team have struggled with different parts of the content of this course at various career stages. We've been there; we're here now to help you have a better experience. We do not want anyone to feel like they are bad at research methods and statistics because they are confused or have difficulty with the software or because it isn't coming easily.

We are here to empower every single one of you!

DETAILED SCHEDULE

The course schedule is subject to change; topic dates are approximate.

Section 1: Research Methods & Statistics Refresh

Week 1	Jan 8	205L Goals; 204L Review		Assignment W0 & W1 (1/11)
Week 2	Jan 13 Jan 15	Scientific manuscripts Group project meetings	Lab W2	Assignment W2 (1/18)

Section 2: Survey Design & Group Differences

Week 3	Jan 20 Jan 22	Survey Design Group project meetings	Lab W3	Assignment W3 (1/25)
Week 4	Jan 27 Jan 29	Experimental study designs Experimental study designs (cont.)	Lab W4	Assignment W4 (2/1)
Week 5	Feb 3 Feb 5	Mean comparison with t-tests Mean comparisons with ANOVA	Lab W5	Assignment W5 (2/8)

Section 3: Factorial Designs and Interactions

Week 6	Feb 10 Feb 12	Midterm prep Midterm prep answer review	Lab W6	Assignment W6 (2/15)
Week 7	Feb 17 Feb 19	Midterm Exam 1 Midterm review in class	Lab W7	Assignment W7 (3/22)
Week 8	Feb 24 Feb 26	Factorial study designs Factorial study designs (cont.)	Lab W8	Assignment W8 (3/1)
Week 9	Mar 3 Mar 5	Interactions in ANOVA Interactions in ANOVA (cont.)	Project meetings	Assignment W9 (3/8)

		Spring Break		
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Week 10	Mar 17 Mar 19	Interactions in regression Group project meetings		Assignment W10 (3/22)
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Section 4: Science Communication

Week 11	Mar 24 Mar 26	Communicating with non-scientists Communicating with scientists	Lab W11	Assignment W11 (3/29)
Week 12	Mar 31 Apr 2	Group project meetings Group project meetings		Assignment W12 (4/5)
Week 13	Apr 7 Apr 9	Science presentations Science presentations		Assignment W13 (4/12)
Week 14	Apr 14 Apr 16	Exam prep Midterm Exam 2		Assignment W14 (4/19)
Week 15	Apr 21	Course review and closing thoughts		Assignment W15 Project report (4/22)

FINAL EXAM on 5/2/26 2-5pm Physics 128